Predicting the Unpredictable: AI Techniques for Supply Chain Risk Mitigation

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Abstract. The past decade has exposed, with uncomfortable regularity, how vulnerable modern supply chains are to disruption. Pandemic-induced shocks, geopolitical instability, and the cumulative fragility of lean, globally distributed value networks have pushed risk management to the centre of operations research and managerial practice alike. In response, artificial intelligence has attracted growing attention as a means of shifting the field away from reactive contingency planning and toward more anticipatory, data-driven approaches to resilience. Yet the body of work at this intersection remains scattered across methodological traditions, risk taxonomies, and industrial sectors, making it difficult to draw coherent conclusions about where progress has genuinely been made and where significant gaps persist. This paper addresses that dispersion through a systematic literature review conducted in accordance with the PRISMA 2020 guidelines. Drawing on searches of the IEEE Xplore and Web of Science databases, and covering publications from 2019 to 2026, the review applies a rigorous two-stage screening protocol to an initial corpus of 192 records, retaining 19 studies for qualitative analysis. The analysis is organised around three research questions: how AI is applied to mitigate supply chain risks, which risk categories are addressed, and which techniques are deployed. The aim is not to advocate for any particular approach but to offer a structured, evidence-based account of the current state of the field and the directions most warranted by what the literature, taken as a whole, reveals.

Keywords: Artificial Intelligence · Supply Chain · Risk Mitigation · Machine Learning

1 Introduction

The past fifteen years have delivered a sequence of disruptions severe enough to force a rethinking of assumptions that had gone largely unquestioned for decades. The 2011 Tohoku earthquake brought automotive and electronics production to a standstill across multiple continents, tracing supply dependencies that few organisations had mapped in full. The COVID-19 pandemic, arriving nearly a decade later, did something similar but at a scale and speed that left little time for adaptation: single-source strategies that had looked efficient suddenly looked reckless, and shortages accumulated faster than contingency plans could address them. The 2021 blockage of the Suez Canal added a different kind of lesson, that a single point of congestion, resolved within days, could translate into delays for hundreds of organisations simultaneously. None of these events were, in hindsight, unforeseeable in category. What they exposed was that the dominant logic of supply chain design, built around leanness, cost optimisation, and the assumption of reasonable stability, had quietly accumulated fragilities that traditional management tools were not built to detect.

The response in both practice and research has been a turn toward resilience, not as a vague aspiration but as a design property, one that requires supply chains to anticipate disruptions rather than simply absorb them. This shift has brought renewed attention to the question of what kinds of tools can actually support that anticipatory capacity. Rule-based systems and classical statistical models were built for environments with stable historical distributions; they tend to underperform precisely when conditions depart most sharply from past experience. Artificial intelligence offers a different kind of capability. Machine learning and deep learning architectures can identify non-linear patterns in high-dimensional, heterogeneous data that would escape conventional analysis. Natural language processing can extract disruption signals from news feeds, regulatory announcements, and social media well before those signals appear in operational data. Graph-based models can represent the relational structure of multi-tier supplier networks and trace how failures propagate through them, something that tabular approaches cannot do without losing the topology that makes propagation possible in the first place.

These capabilities have attracted considerable research interest, and the empirical literature has grown accordingly. The difficulty is that it has grown in several directions at once, across methodological traditions, risk taxonomies, industrial sectors, and geographic contexts that do not always speak to one another. Identifying what the field actually knows, as opposed to what individual studies demonstrate under particular conditions, requires a more systematic effort of synthesis than the primary literature alone can provide.

This paper undertakes that synthesis through a systematic literature review conducted in accordance with the PRISMA 2020 framework. Starting from 192 records retrieved from IEEE Xplore and Web of Science, and covering publications from 2019 to 2026, the review applies a two-stage screening protocol to arrive at 19 studies retained for qualitative analysis. Three research questions organise the work: (RQ1) how AI is applied to mitigate supply chain risks; (RQ2) which risk types are addressed; and (RQ3) which AI techniques are employed. The analysis goes beyond descriptive mapping to identify structural gaps, geographic concentration in Asian research contexts, limited coverage of cybersecurity and geopolitical risks, near- absent integration of real-time data infrastructures, that point toward the directions where the field most needs to develop.

The remainder of this paper is organised as follows. Section 2 establishes the theoretical background. Section 3 describes the methodology. Sections 4 and 5 present the selection results and content analysis. Section 6 discusses key findings, emerging trends, and identified gaps. Section 7 concludes.

2 Theoretical Background

This section establishes the conceptual foundations on which the review rests, covering the contemporary challenges of supply chain management, the main risk categories recurring across the literature, and the AI techniques most relevant to the studies examined.

2.1 Supply Chain Management: Scope and Contemporary Challenges

Supply Chain Management (SCM) encompasses the coordination of activities involved in sourcing raw materials, transforming them into finished goods, and delivering them to end customers, together with the management of material, information, and financial flows that support these processes. For much of the 2000s, lean management was the dominant paradigm: reducing inventory buffers, rationalising supplier bases, and optimising for cost. These principles delivered genuine efficiency gains but created systemic fragility that went largely unnoticed until disruptions struck at scale.

The COVID-19 pandemic exposed this fragility with particular severity. Manufacturing firms dependent on single-source strategies found themselves unable to absorb even brief supply shocks, with shortages persisting well into 2021. Similar dynamics had emerged after the 2011 Tohoku earthquake and the 2021 Suez Canal blockage. These repeated crises accelerated a conceptual shift already visible in the academic literature, with researchers arguing that supply chains need to be designed not merely for efficiency but for *resilience*, the capacity to anticipate disruptions, absorb their impact, and recover within a reasonable timeframe, and, in the case of prolonged crises, for *viability*, meaning the capacity to adapt structurally rather than simply return to a prior steady state.

The globalisation of supply chains has also introduced a complexity that compounds these vulnerabilities. Modern value networks typically span multiple supplier tiers, logistics intermediaries, and regulatory environments across continents. Interdependencies between nodes mean that a disruption at one point can propagate rapidly through the network, the so-called *ripple effect*, making containment one of the central problems to which AI-based approaches are being applied.

2.2 Supply Chain Risk: Categories and Taxonomy

Supply chain risk is broadly understood as the potential for an event to disrupt material, information, or financial flows in ways that harm organisational performance. This review adopts a multi-category taxonomy drawing on the typologies most frequently appearing across the included studies.

Supplier risk refers to the possibility that a supplier will fail to deliver inputs on time, at the required quantity, or quality level, and encompasses financial instability, capacity constraints, and geographic concentration. **Demand risk**

arises from uncertainty in customer orders, including the well-documented bullwhip effect whereby small retail fluctuations are amplified into large upstream production swings. **Disruption risk** covers low-frequency, high-impact external events, pandemics, natural disasters, geopolitical conflicts, port closures, that are poorly represented in historical data. **Operational risk** encompasses internally originating failures such as equipment breakdowns, process errors, and IT system failures. **Financial and fraud risk**, including payment defaults and invoice manipulation, receives comparatively less attention in the corpus but is identified as a clear use case for explainable AI. Two further categories, environmental and sustainability risk and cybersecurity risk, are of growing prominence in the broader literature, though they remain underrepresented in this corpus.

2.3 Artificial Intelligence in Supply Chain Management

Artificial intelligence, in the context of this review, denotes computational methods that enable systems to learn from data, detect patterns, and support or automate decisions. The corpus reflects a broad methodological spectrum.

Traditional machine learning, including Random Forest, Support Vector Machines, and gradient boosting methods such as XGBoost and LightGBM, remains the most prevalent approach, valued for its robustness on structured tabular data and its interpretability relative to deeper architectures. Unsupervised methods, particularly K-Means clustering, are applied to supplier segmentation and anomaly detection.

Deep learning architectures, foremost Long Short-Term Memory networks (LSTM) and Gated Recurrent Units (GRU), are applied to demand forecasting, disruption prediction, and time-series anomaly detection. Hybrid architectures combining LSTMs with Graph Neural Networks (GNNs) model both the temporal evolution of risk indices and the structural relationships between supply chain nodes. One notable study develops a sequence-to-disruption architecture integrating bidirectional GRUs with transformer-style attention, achieving prediction errors under 20 days for six-month-ahead shortage forecasts on a dataset of 10,000 automotive components.

Graph Neural Networks operate on graph-structured supply chain data, learning how disruptions propagate through network topology. One reviewed framework constructs a supply chain knowledge graph with over 2.5 million entities and eleven relationship types to support real-time risk monitoring and recovery decisions.

Natural language processing enables structured information to be extracted from unstructured text. Applications include sentiment analysis of social media streams for real-time early warning and named entity recognition pipelines applied to longitudinal news archives for risk categorisation.

Explainable AI (XAI), particularly SHAP (SHapley Additive Explanations), addresses the interpretability deficit of complex models. Its adoption reflects a practical imperative: supply chain risk decisions carry significant operational and financial consequences, requiring not only accurate predictions but transparent justifications that managers can interrogate and audit.

Federated machine learning addresses the competitive and regulatory barriers to centralising sensitive supplier performance data by allowing organisations to collaboratively train shared risk models without raw data leaving individual servers. Studies in the corpus indicate that federated models achieve performance close to centralised counterparts while preserving data privacy, a trade-off particularly attractive for small and medium-sized enterprises.

3 Methodology

This section describes the systematic literature review (SLR) protocol adopted to identify, screen, and synthesise the body of evidence concerning the use of Artificial Intelligence (AI) for risk mitigation in Supply Chain Management (SCM). The procedure was designed to ensure transparency, reproducibility, and methodological rigour throughout every stage of the review.

3.1 Review Protocol

The review followed the *Preferred Reporting Items for Systematic Reviews and Meta-Analyses* (PRISMA) 2020 guidelines, which provide a structured framework for reporting the identification, screening, and inclusion of studies in systematic reviews.

The overarching goal of the protocol was to answer, in a systematic and reproducible manner, the following research questions (RQs):

- **RQ1:** How can Artificial Intelligence be applied to mitigate risks in supply chains?
- **RQ2:** Which types of supply chain risks are addressed through AI-based approaches?
- **RQ3:** Which Artificial Intelligence techniques are employed for supply chain risk mitigation?

Each phase of the review—identification, screening, eligibility, and inclusion—was documented in accordance with the PRISMA 2020 flow diagram, allowing the entire selection process to be audited and replicated.

3.2 Databases Searched

Two scientific databases were selected as primary sources for the literature search: **IEEE Xplore** and **Web of Science**.

The coverage period was restricted to publications released between **2019 and 2026**. This window was chosen to capture the most recent wave of AI adoption in SCM, including the period following major disruptions such as the COVID-19 pandemic and subsequent geopolitical shocks, which have substantially reshaped the academic discourse on supply chain resilience and risk.

3.3 Search Queries

The search string was constructed along three conceptual dimensions: (i) the application *domain* (Supply Chain), (ii) the *technology* of interest (Artificial Intelligence and its synonyms), and (iii) the *type of analysis* pursued (Risk Mitigation). Boolean operators were used to combine the dimensions, while quotation marks ensured exact-phrase matching. The final query is presented below:

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"Supply Chain" AND ("Artificial Intelligence" OR "AI") AND "Risk
Mitigation"
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The query was applied to the title, abstract, and keyword fields of both databases. Table 1 summarises the number of records retrieved per database.

Table 1. Number of records retrieved per database.

Database	**Records Retrieved**
IEEE Xplore	142
Web of Science	50
Total	**192**

The rationale behind combining these three dimensions was to balance recall (broad enough to capture different AI families, e.g., Machine Learning, Deep Learning, NLP) with *precision* (narrow enough to filter out studies that, while mentioning AI or SCM, do not address the risk dimension explicitly).

3.4 Screening Criteria

A first set of thematic filters was applied during the screening phase, that is, while reading the title, keywords, and abstract of each record. To advance beyond this stage, an article had to simultaneously satisfy the following three requirements:

1. **Exclusive Supply Chain context.** The study had to demonstrate a practical application within logistics, procurement, inventory management, production planning, or transport networks. Articles addressing supporting technologies such as Blockchain or IoT from a purely technical perspective (e.g., network architecture or communication protocols), without applying them to logistics operations, were rejected.
2. **Explicit presence of Artificial Intelligence.** Studies were required to employ advanced AI techniques, such as Machine Learning (ML), Neural Networks, Deep Learning (DL), Natural Language Processing (NLP), or genetic algorithms. Works limited to basic digitalisation, traditional automation, ERP systems, or the mere deployment of IoT/Blockchain sensors for data collection—without AI-based predictive or prescriptive processing—were excluded.

3. **Central focus on risk prevention and mitigation.** The AI component had to be deployed in order to anticipate disruptions, manage market uncertainty (supply or demand), build resilience, foresee supplier failures, or minimise the impact of adverse events. Applications targeting strictly operational or commercial goals (e.g., optimising daily delivery routes or sales forecasting for marketing purposes), without a clear risk or resilience dimension, were excluded.

3.5 Eligibility Criteria

Articles that passed the screening phase were then subjected to a more demanding set of inclusion and exclusion criteria, applied during the full-text reading. The objective at this stage was no longer to identify articles broadly related to AI and Supply Chain, but to retain only studies offering a *direct, substantive, and methodologically explicit* contribution to the research questions formulated in Section 3.1.

Inclusion Criteria. An article was retained only if it satisfied *all* of the following requirements:

- **IC1 — Centrality of AI:** Artificial Intelligence had to be a central element of the article, rather than a peripheral or merely cited concept.
- **IC2 — Risk-oriented purpose:** The main focus of the study had to be the mitigation, prediction, management, or assessment of risk in a supply chain context.
- **IC3 — Methodological clarity:** The article had to describe explicitly (i) the AI techniques employed, (ii) the type(s) of risk addressed, and (iii) a concrete application, model, or methodology.
- **IC4 — Tangible contribution:** The work had to present results, a framework, a model, an empirical study, or an applicable methodology.
- **IC5 — Alignment with the RQs:** The article had to provide an explicit contribution to at least one of the three research questions (RQ1, RQ2, or RQ3).
- **IC6 — Generalisability:** The context studied had to be sufficiently transferable to business and commercial supply chain settings.

Exclusion Criteria. Conversely, an article was excluded whenever *any* of the following situations occurred:

- **EC1 — Superficial use of AI:** AI mentioned without an underlying implementation or concrete methodology.
- **EC2 — Operational focus without risk:** Primary emphasis on optimisation, efficiency, logistics, or operational performance with no clear link to risk mitigation.

– **EC3 — AI as an accessory technology:** The article centres on digital transformation, Industry 4.0, IoT, Blockchain, or Big Data as its main theme, with AI playing only an ancillary role.
– **EC4 — Unclear risk typology:** Failure to clearly identify the type(s) of supply chain risk being addressed.
– **EC5 — Absence of a specific AI technique:** No specific AI model, algorithm, or method is described.
– **EC6 — Excessive abstraction:** Overly conceptual or generic articles, lacking applicable results.
– **EC7 — Out-of-scope risk:** The risk discussed is not genuinely a supply chain risk.
– **EC8 — Limited transferability:** Studies confined to highly idiosyncratic contexts, whose findings cannot reasonably be generalised to broader business settings.

3.6 Screening Process

The selection of studies was organised in three sequential phases, in line with the PRISMA 2020 flow.

Phase 1 — Identification and removal of duplicates. The 192 records exported from IEEE Xplore and Web of Science were consolidated into a single reference manager. Three duplicate entries were identified and removed, resulting in 189 unique records advancing to the next stage.

Phase 2 — Title, keywords, and abstract screening. Each remaining record was evaluated based on its title, keywords, and abstract, against the three thematic filters defined in Section 3.4. All three filters had to be simultaneously satisfied for an article to advance to the next phase. To ensure transparency and traceability, this stage was fully documented in a dedicated spreadsheet, in which each record was registered together with the following fields: *Title*, *Decision*, *Risk Analysed*, *AI Technique*, and *Justification*. This structure made it possible to revisit individual decisions, audit the rationale behind each exclusion, and verify the consistency of judgements across the dataset. At the end of this phase, 128 records were removed (23 from IEEE Xplore and 105 from Web of Science), leaving 61 studies eligible for full-text assessment.

Phase 3 — Full-text reading and eligibility assessment. The 61 surviving articles were retrieved in full and subjected to a stricter evaluation against the inclusion and exclusion criteria defined in Section 3.5. This phase was also documented in a structured spreadsheet, where each article was characterised through a richer set of descriptors aimed at supporting both the eligibility decision and the subsequent qualitative synthesis. For every record the following fields were registered: *Title*, *Authors*, *Year*, *Journal/Conference*, *Country of the Authors*, *Inclusion Criteria Satisfied*, *Exclusion Criteria Triggered*, *Decision*, *Justification of the Decision*, *AI Techniques*, *Types of Risk Addressed*, *Mitigation Approach*, *Application Sector*, *Type of Study*, *Answer to RQ1*, *Answer to RQ2*, *Answer to RQ3*, *Problem Addressed*, *Main Contribution*, *Limitations*, and *Future Work*.

This extended template fulfilled a dual purpose. On the one hand, it operationalised the eligibility assessment by forcing an explicit mapping between each article and the inclusion and exclusion criteria, as well as the perceived strength of relevance (high, medium, or low). On the other hand, it already constituted the basis for the data extraction stage, since the AI techniques, risk types, mitigation approaches, and contributions per research question would later feed directly into the qualitative synthesis. A conservative posture was deliberately adopted: in cases of doubt, or whenever the methodological description of either the AI component or the risk dimension was insufficient, the article was excluded. Studies classified as having only *low* relevance were likewise discarded, even when formally meeting the basic inclusion criteria.

Of the 61 articles assessed in this phase, 2 could not be retrieved in full text and 40 were excluded for failing to meet at least one of the criteria above, resulting in a final corpus of **19 studies** included in the qualitative synthesis.

Reviewer procedure. The screening and eligibility assessment were conducted as a team effort, with the set of articles distributed among the authors and evaluated individually against the criteria defined above. Each article was therefore reviewed by a single author, but the criteria, the spreadsheet templates, and the decision rubric were collectively defined and applied uniformly across the team. Borderline cases and decisions flagged as uncertain were brought to joint discussion and re-examined collectively, in order to ensure consistency of judgement across reviewers and to mitigate individual bias.

Figure 1 summarises the full selection process, from the 192 records initially identified to the 19 studies retained for analysis.

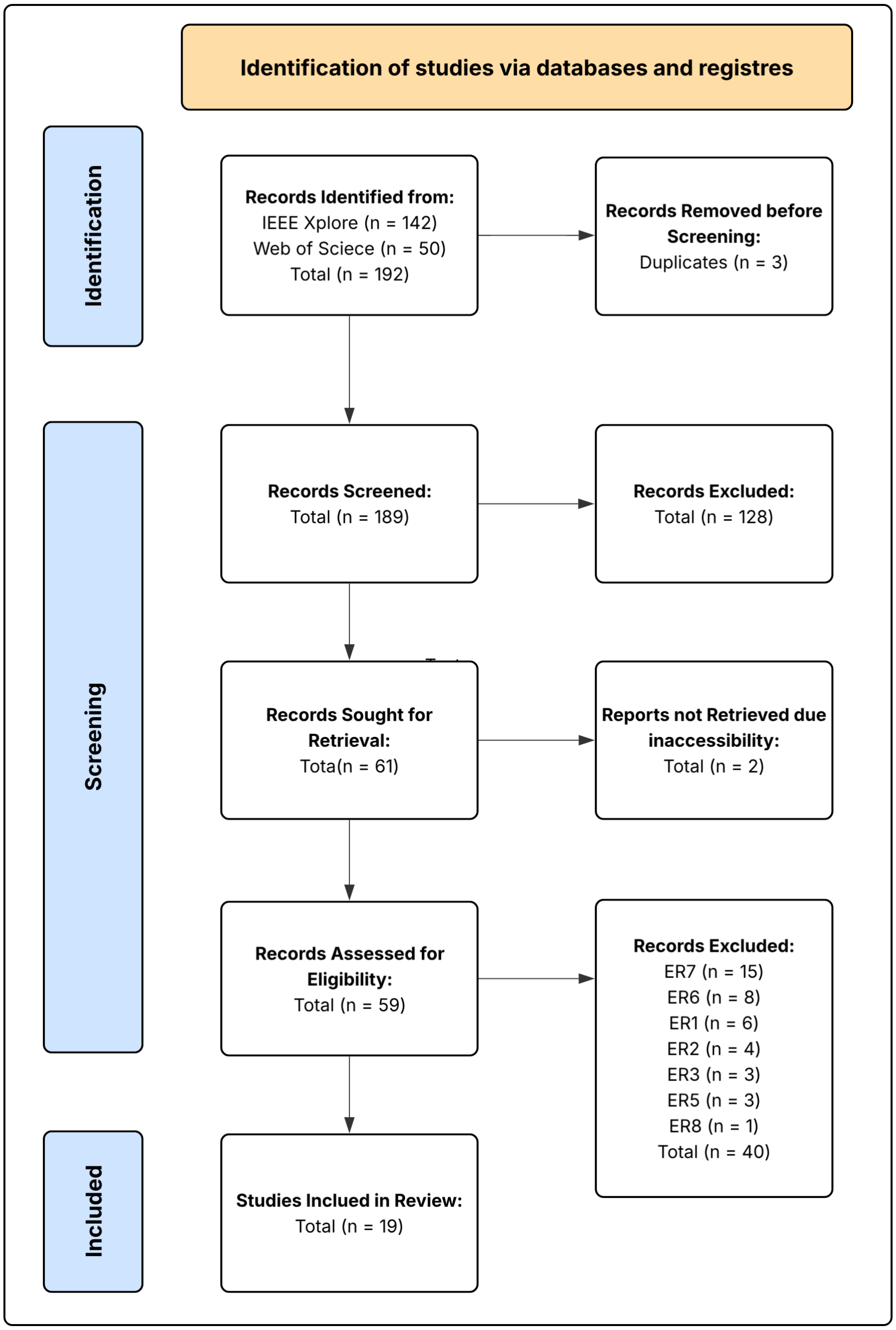

Fig. 1. PRISMA 2020 flow diagram of the study selection process.

4 Selection Results

Before addressing the research questions, a descriptive characterization of the 19 selected articles is presented. This analysis contextualizes the state of the field by examining the temporal evolution of publications, geographic concentration, methodological diversity, and sectoral coverage of the included studies.

4.1 Temporal Distribution

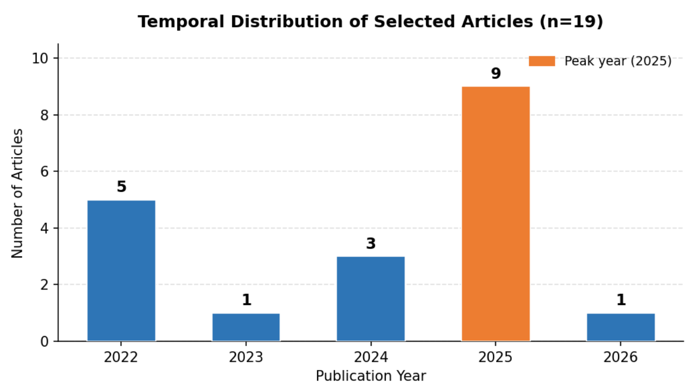

Fig. 2. Temporal distribution of selected articles (n=19).

The temporal distribution of the 19 selected articles (Fig. 2) reveals a clear acceleration in research activity at the intersection of Artificial Intelligence and Supply Chain Risk Management (SCRM) over recent years. The earliest cluster (2022) accounts for five publications, coinciding with a period of heightened academic interest driven by the operational disruptions caused by the COVID-19 pandemic and subsequent supply chain instability. A relative consolidation is observed in 2023 (n=1) and 2024 (n=3), suggesting a transitional period as the field matured and pivoted towards more technically sophisticated approaches.

The most notable finding is the sharp surge in 2025, which alone concentrates nine of the nineteen articles (47.4%), representing the single most productive year in the corpus. This trajectory strongly indicates that the application of AI to SCRM is an emergent and rapidly evolving domain, where foundational conceptual work (2022) is giving way to empirical validation and system deployment (2025–2026). The presence of one article from 2026 further corroborates that the research frontier is active and ongoing, underscoring the timeliness and relevance of this systematic review.

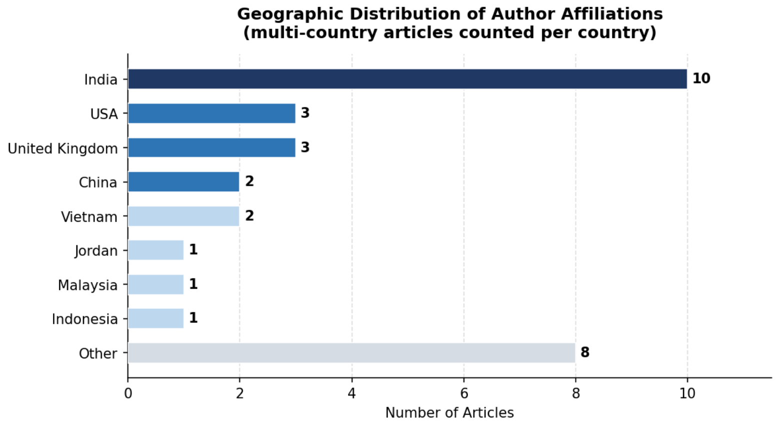

Fig. 3. Geographic distribution of author affiliations (multi-country articles counted per country, n=19).

4.2 Geographic Distribution

The geographic analysis (Fig. 3) reveals a pronounced concentration of research output originating from Asia and Anglo-Saxon academic contexts. India emerges as the dominant contributor, with affiliations appearing in ten of the nineteen articles, a figure that reflects both the country's substantial investment in AI research capacity and the strategic importance of supply chain resilience in its manufacturing and logistics sectors. The United States (n=3) and United Kingdom (n=3) follow, consistent with their historically strong publication output in operations management and computer science. China (n=2) and Vietnam (n=2) are also represented, both nations with significant stakes in global manufacturing networks.

A critical observation is the near-complete absence of European and South American authorship. With the exception of one article from Germany and one from Hungary (both as co-authors in multi-country collaborations), the European research community appears largely underrepresented in this specific intersection of AI and SCRM. This constitutes a meaningful geographic gap, as European supply chains—heavily regulated and deeply integrated through European Union frameworks—may face distinct risk profiles not adequately captured by models developed in other geopolitical and industrial contexts. Future research should prioritize geographically diverse empirical studies to ensure generalizability.

4.3 Study Type and Methodological Profile

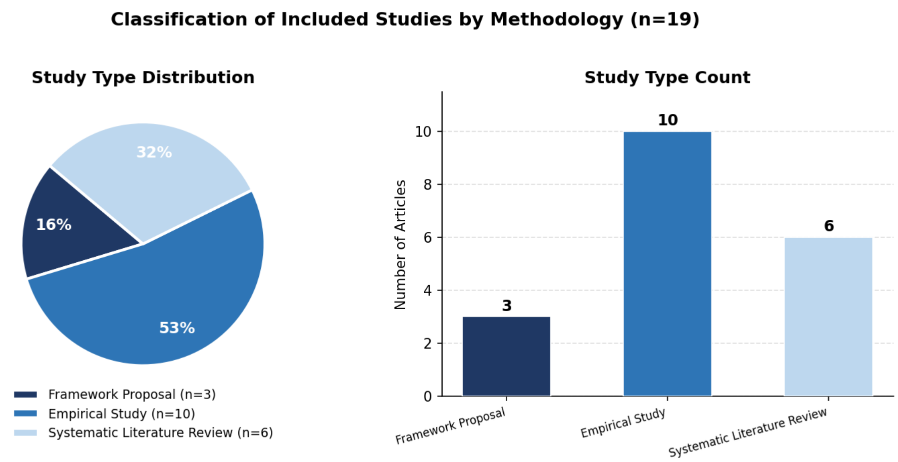

Fig. 4. Classification of included studies by methodology (n=19).

The methodological profile of the corpus (Fig. 4) is dominated by *Empirical Studies* (n=10, 52.6%), which encompass case studies, computational experiments with real or simulated datasets, and validation studies applying AI models to supply chain data. This predominance signals that the field has progressed beyond purely theoretical discussion and is actively engaged in operationalizing AI solutions in applied settings.

Systematic Literature Reviews (SLR) constitute the second largest group (n=6, 31.6%), including bibliometric analyses and state-of-the-art reviews. The presence of six SLRs within a 19-article corpus is noteworthy, as it suggests that the community itself recognizes the need for synthesis and consolidation—a signal that the field is reaching a level of maturity where meta-analytical work is valued alongside primary research. This also implies that the knowledge base, while growing rapidly, remains fragmented, warranting ongoing integrative efforts such as the present review.

Framework Proposals (n=3, 15.8%) round out the corpus, representing conceptual contributions that map AI architectures and processes for SCRM without full empirical validation. These works serve as design blueprints for future implementations but highlight a gap: further empirical testing of proposed frameworks in real-world, longitudinal supply chain settings remains scarce.

4.4 Sectoral Coverage

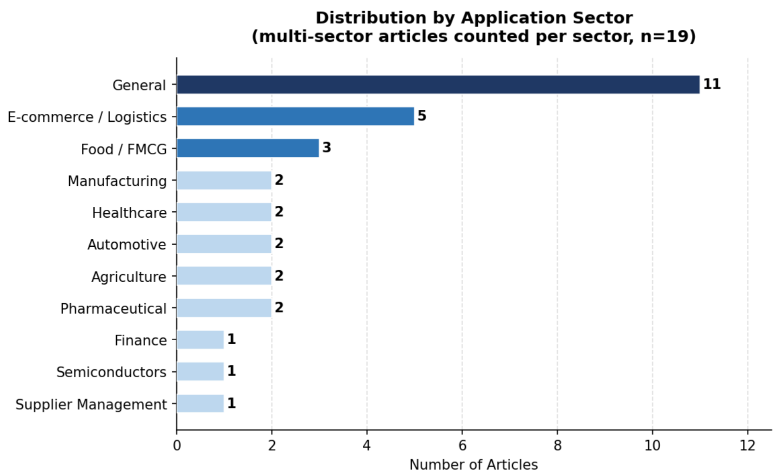

Fig. 5. Distribution of articles by application sector (multi-sector articles counted per sector, n=19).

The sectoral analysis (Fig. 5) exposes a significant limitation in the current literature: the majority of articles (n=11) adopt a *general* scope, applying AI-based SCRM frameworks without targeting a specific industry vertical. While such generality offers broad applicability, it simultaneously limits practical guidance for practitioners operating within sectors subject to idiosyncratic regulatory, operational, and risk environments.

Among sector-specific contributions, *E-commerce and Logistics* (n=5) represents the most investigated context, reflecting the real-time data availability and transaction volume that make this sector particularly well-suited to AI-driven risk detection. *Food and FMCG*, *Healthcare*, *Automotive*, *Agriculture*, and *Pharmaceuticals* each appear in two articles, indicating nascent but emerging interest. Notably, sectors such as *Financial Services*, *Semiconductors*, and *Energy*—all of which face severe and well-documented supply chain vulnerabilities—are critically underrepresented.

This sectoral breadth gap is a substantive finding of this review. It implies that the AI techniques currently proposed may not be calibrated to sector-specific risk taxonomies, compliance constraints, or data infrastructures, and that domain-adapted AI models represent a priority avenue for future research.

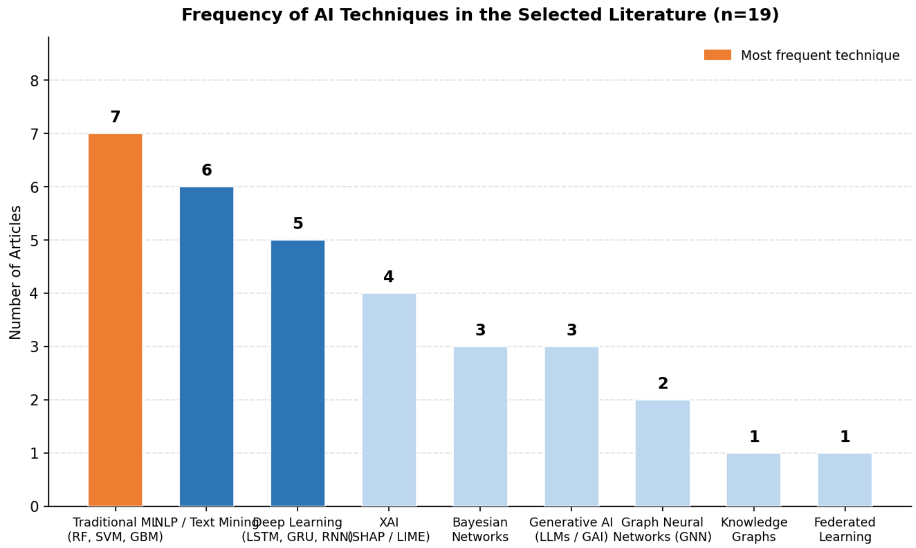

Fig. 6. Frequency of AI techniques identified across the selected literature (n=19).

4.5 Frequency of AI Techniques

The frequency analysis of AI techniques employed across the corpus (Fig. 6) reveals a clear hierarchy in adoption, with established, interpretable methods predominating over state-of-the-art generative or graph-based approaches.

Traditional Machine Learning (RF, SVM, GBM, and variants) is the most prevalent technique, appearing in seven articles. Its dominance is attributable to its strong performance on structured tabular data—the dominant format in ERP, WMS, and supplier management systems—as well as its relative computational efficiency and interpretability compared to deeper architectures. *NLP and Text Mining* (n=6) constitutes the second most frequent category, reflecting the growing recognition that unstructured data sources—social media, news feeds, regulatory announcements—contain early signals of supply chain stress that structured internal data cannot capture.

Deep Learning architectures (LSTM, GRU, RNN; n=5) rank third, used primarily for sequential and time-series forecasting tasks. *Explainable AI* (SHAP, LIME; n=4) has gained meaningful traction, consistent with a growing demand for algorithmic transparency in high-stakes operational contexts. *Bayesian Networks* (n=3) and *Generative AI* (LLMs, GAI; n=3) appear with equal frequency, although their use cases differ markedly: the former for probabilistic multi-factor risk assessment, the latter for processing unstructured disruption signals.

At the lower end of adoption, *Graph Neural Networks* (n=2), *Knowledge Graphs* (n=1), and *Federated Learning* (n=1) remain niche despite their theoretical promise for modelling interconnected supply network topologies and privacy-preserving collaborative risk management, respectively. This underrepresentation is a pivotal finding: techniques most suited to the networked, relational nature of supply chains are the least deployed, suggesting that the field has not yet fully leveraged the structural characteristics of its own domain.

5 Content Analysis

5.1 Answer to RQ1 — How AI Mitigates Supply Chain Risks

The analysis of the 19 included articles reveals six distinct mechanisms through which artificial intelligence is applied to mitigate risks in supply chains. Early warning and forecasting constitutes the dominant approach, identified in eight articles (42%), followed by automation and decision support and resilience and recovery, each present in four articles (21%). Privacy-preserving collaboration, explainability and transparency (XAI), and conceptual framework or review contributions each account for one article apiece, together representing the remaining 16% of the corpus.

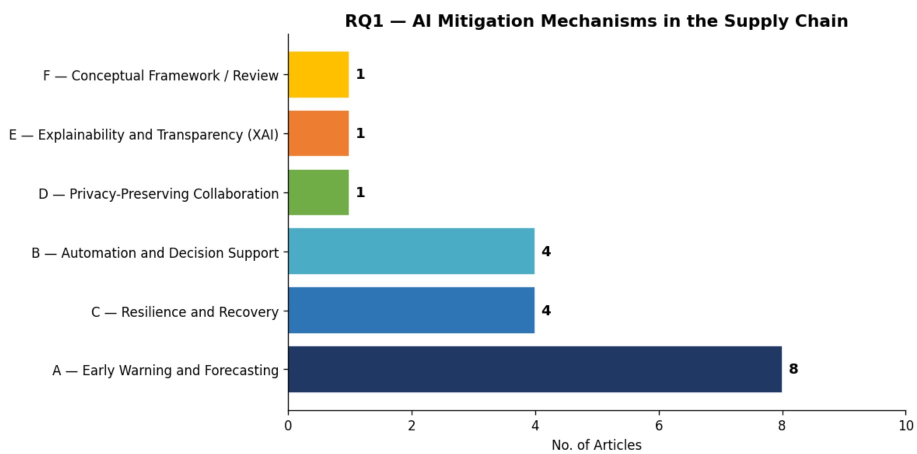

Fig. 7. Distribution of AI mitigation mechanisms across the 19 included articles (RQ1).

Early warning and forecasting is the most pervasive mitigation mechanism in the literature, reflecting a theoretical consensus that preemptive anticipation of supply chain disruptions is more effective than post-event recovery. Studies in this category deploy AI models trained on historical operational and environmental data to generate alerts before risks materialise. Gradient boosting architectures (including LightGBM, XGBoost, and Random Forest) are applied to predict delayed deliveries and payment anomalies [1], while deep learning frameworks combining GRU units, Bidirectional RNNs, and Transformer layers translate raw organisational data into probabilistic disruption forecasts [10]. Deep

Reinforcement Learning combined with Graph Neural Networks and Bayesian Networks enables proactive routing and adaptive anticipation of systemic disruptions and supplier failures [17]. Hybrid GNN-LSTM models capture both spatial supplier-relationship embeddings and temporal risk patterns simultaneously [5]. Machine learning and data mining underpin a collaborative early warning system for China's manufacturing sector [18], and knowledge graph-based frameworks integrate Graph Neural Networks and weakly supervised learning for risk monitoring, early warning, and decision support [19]. Real-time text mining and sentiment analysis applied to social media streams extend the early warning paradigm to unstructured data sources, enabling detection of emerging disruptions such as port congestion and semiconductor shortages as they develop [4]. Review contributions document the broader transition from reactive to proactive risk management enabled by this class of AI application [3, 9].

Automation and decision support encompasses AI-driven systems that reduce human workload in risk-related decisions or automate responses to identified threats. Lin et al. [11] introduce the UNISONE framework, which combines a Generative AI module for extracting disruption signals from unstructured text with Markov state modelling and Mixed Integer Programming for adaptive capacity reallocation. Explainable AI methods incorporating SHAP attribution and TOPSIS multi-criteria ranking automate and transparently support supplier selection decisions [8]. A multi-agent architecture integrating an LLM, an LSTM autoencoder, Random Forest, K-Means clustering, and Reinforcement Learning automates routing, anomaly detection, ESG scoring, and risk classification within a unified pipeline [16]. Clustering algorithms that segment suppliers into risk-performance profiles based on multiple operational indicators further exemplify AI-driven automation of vendor management decisions [7].

Resilience and recovery captures studies that use AI to model, simulate, or support the restoration of supply chain continuity following disruption. Fernando et al. [6] examine how machine learning informs manufacturing firms' adaptive responses to COVID-19 disruption, while Nayal et al. [12] investigate how AI adoption supports the agro-industry in sustaining operations under pandemic-induced supply chain stress. Nguyen et al. [14] demonstrate how Federated Machine Learning enables collaborative risk prediction across supply chain partners to support network-level resilience, and Naz et al. [13] systematically document AI as an enabler of supply chain resiliency in the post-COVID-19 context.

Privacy-preserving collaboration is addressed in one article. Nguyen et al. [14] show that Federated Machine Learning allows supply chain partners to collaboratively train risk prediction models (combining CNN and LSTM components) without sharing proprietary raw data, resolving a fundamental barrier to multi-stakeholder AI adoption.

Explainability and transparency (XAI) as a standalone mitigation mechanism is the focus of one systematic review. Nimmy et al. [15] comprehensively evaluate the suitability of XAI algorithms for supply chain operational risk management, establishing a methodological benchmark for transparent risk detection and positioning explainability as a first-class design requirement for applied AI in this domain.

Conceptual framework and review contributions provide the theoretical scaffolding on which applied studies build. Aldehayyat et al. [2] propose a digital transformation framework integrating AI and machine learning within an agile management paradigm, addressing the organisational prerequisites for AI-driven risk mitigation.

Overall, the dominance of early warning and forecasting reflects an underlying conviction in the literature that the most effective form of supply chain risk management is pre-emptive. The emergence of multi-agent and LLM-based architectures in 2025 articles suggests that the field is beginning to operationalise this logic at higher temporal resolution, moving from batch prediction toward continuous, automated risk surveillance.

5.2 Answer to RQ2 — Types of Supply Chain Risks Addressed

The cross-reference matrix produced from the 19 included articles reveals eight distinct risk type categories addressed through AI-based approaches: supplier risk, disruption risk, logistics and transport risk, demand and inventory risk, cybersecurity risk, pandemic risk, geopolitical risk, and fraud and integrity risk. Coverage is unequal across these categories, with logistics and transport risk and disruption risk attracting the highest number of AI technique applications, while cybersecurity, geopolitical, and fraud risks remain markedly under-represented.

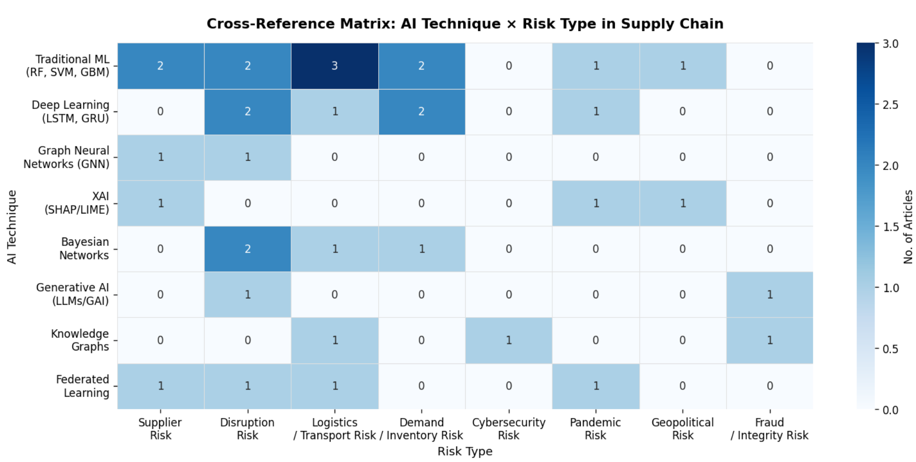

Fig. 8. Cross-reference matrix of AI techniques and supply chain risk types addressed in the included articles. Cell values indicate the number of articles addressing each combination. Empty cells indicate no article in the corpus addresses that particular technique-risk pairing.

Logistics and transport risk (encompassing transportation disruptions, port congestion, lead time variability, and distribution network failures) receives the broadest AI coverage across the corpus. Traditional machine learning methods (Random Forest, SVM, GBM) address this category in three articles, with additional coverage from Deep Learning, Bayesian Networks, Knowledge Graphs, and Federated Learning, reflecting the data richness and operational measurability of logistics processes [4, 19, 14].

Disruption risk is addressed by the widest range of AI technique categories (traditional ML, Deep Learning, Bayesian Networks, Generative AI, and Federated Learning), confirming it as the most cross-cutting risk type in the literature. This breadth reflects the scale of academic response to the COVID-19 pandemic and subsequent geopolitical instability, which exposed the fragility of global supply networks to macro-level shocks [10, 11, 14, 6].

Supplier risk (covering supplier failure, quality issues, delivery delays, and credit default) is addressed predominantly through traditional ML methods and GNNs, which are particularly suited to modelling network-level interdependencies among suppliers [5, 19, 8, 17].

Demand and inventory risk, arising from demand volatility, forecast errors, and inventory misalignment, is addressed by traditional ML and Deep Learning in two articles each, reflecting the long-standing application of statistical learning to demand forecasting problems [1, 10].

Pandemic risk receives dedicated attention in three articles through traditional ML, XAI, and Federated Learning, confirming the COVID-19 pandemic as a structuring event for this research stream [12, 15, 14].

Geopolitical risk is addressed in two articles through traditional ML and XAI-based approaches [2, 8]. *Fraud and integrity risk* appears in three articles: Generative AI and Knowledge Graphs are applied to fraud detection in supply chain operations [11, 19], while Federated Machine Learning is specifically deployed for food fraud detection in collaborative supply chain environments [14]. *Cybersecurity risk* is the least studied category in the corpus, addressed in two articles through Knowledge Graph-based and Federated Learning approaches [19, 14], a finding that points to a significant gap in a field where supply chain digitalisation is accelerating.

5.3 Answer to RQ3 — AI Techniques Used

The methodological landscape of the 19 included articles encompasses eight distinct AI technique categories. Traditional machine learning (comprising Random Forest, Support Vector Machines, and Gradient Boosting Methods) is the most widely deployed category, applied across the broadest range of risk types. Deep Learning (LSTM, GRU), Graph Neural Networks (GNN), XAI (SHAP/LIME), Bayesian Networks, Generative AI (LLMs), Knowledge Graphs, and Federated Learning constitute the remaining categories, reflecting a methodologically diverse corpus.

Traditional machine learning (Random Forest, SVM, GBM) demonstrates the broadest risk coverage of any single technique category, addressing supplier, disruption, logistics and transport, demand and inventory, pandemic, and geopolitical risks across the corpus [1, 2, 18, 7]. The versatility of these methods for structured, tabular supply chain data explains their continued prevalence despite the emergence of more complex architectures.

Deep Learning (LSTM, GRU) is applied to disruption, logistics and transport, and demand and inventory risks, with recurrent architectures particularly suited to supply chain time-series data where historical sequences encode the dynamics of disruption propagation [10, 5]. Deep Reinforcement Learning extends this paradigm further, enabling adaptive routing decisions in response to systemic disruptions and supplier failures [17].

Graph Neural Networks address both supplier risk and logistics and transport risk [19, 5, 17], reflecting the natural fit between graph-structured models and the network topology of supply chains, where risks propagate through supplier-buyer relationships.

XAI (SHAP/LIME) is applied across supplier, pandemic, and geopolitical risks, confirming that high-stakes, low-frequency risk decisions create the strongest demand for interpretable AI outputs [8, 15, 10, 1].

Bayesian Networks address disruption, logistics and transport, and demand and inventory risks, providing probabilistic modelling capabilities well-suited to uncertainty quantification under incomplete information [3, 12, 17].

Natural Language Processing (NLP) is applied to supply chain risk identification and early warning through text mining, sentiment analysis, and association rule mining on social media and operational data streams [4, 13, 18, 16, 11]. These approaches enable the extraction of risk signals from unstructured data sources that structured ML models cannot directly process.

Generative AI (LLMs) appears in two articles addressing disruption risk and fraud and integrity risk [11, 16], representing the most recent technical development in the corpus and signalling the entry of large language model capabilities into supply chain risk management.

Knowledge Graphs address logistics and transport risk, cybersecurity risk, and fraud and integrity risk [19], offering structured knowledge representation capabilities for complex, multi-entity risk environments.

Federated Learning addresses supplier, disruption, logistics and transport, and pandemic risks within a single architectural contribution [14], enabling privacy-preserving collaborative risk modelling across organisational boundaries.

A temporal trend is discernible in the distribution of techniques across the corpus. Of the six articles published in 2022 and 2023, three rely primarily on traditional ML or review-based survey methods, while three already deploy hybrid or multi-technique architectures combining NLP, XAI, and statistical modelling. By contrast, of the thirteen articles published between 2024 and 2026, ten deploy hybrid or advanced architectures combining Deep Learning, GNNs, Generative AI, LLMs, or formal optimisation methods, with only three relying on single-technique approaches. This shift reflects both the maturation of these techniques and the increasing availability of the large, diverse datasets that complex architectures require.

5.4 Complementary Analysis

Sector concentration. The majority of included articles propose general or sector-agnostic supply chain models, limiting the ability to draw sector-specific conclusions. Where specific industries are targeted, manufacturing receives the most attention [18, 10], followed by logistics and transportation [9, 16], agriculture [12], e-commerce and contract logistics [1], and multi-sector contexts including pharmaceuticals, food, and semiconductors [11]. Fernando et al. [6] additionally cover manufacturing, food, and healthcare supply chains within their pandemic-focused review. Energy and public sector supply chains remain entirely absent from the corpus, representing areas where AI-based risk management frameworks remain underdeveloped.

Geographic distribution. India is the most represented country in the corpus, with ten articles affiliated with Indian institutions, either solely or in multi-country collaborations [17, 2–5, 7, 8, 12, 13, 16]. China contributes two articles [19, 18] and Vietnam two articles [9, 14]. Further contributions originate from Australia [15], Taiwan and Japan [11], Malaysia and Indonesia [6], and a multi-country collaboration spanning the United Kingdom, Ghana, and the United States [1]. Li et al. [10] represent a collaboration across the United States, United Kingdom, and Germany. Primary contributions from North American and continental European institutions as lead affiliations are largely absent, despite these regions' substantial supply chain research traditions.

Temporal evolution. The corpus spans the period from 2022 to 2026, with five articles published in 2022, one in 2023, three in 2024, nine in 2025, and one in 2026, confirming a clear upward trajectory in publication volume. The COVID-19 pandemic serves as the principal inflection point, with the most technically sophisticated contributions (deploying LLMs, multi-agent systems, and advanced hybrid architectures) concentrated in 2025 and 2026.

6 Discussion

6.1 Synthesis of Key Findings

Collectively, the three research questions reveal a field in active and accelerating development, characterised by increasing methodological sophistication and a clear orientation toward preemptive risk management over reactive recovery. Early warning and forecasting emerges as the most frequently pursued application of AI in supply chain risk management, present in approximately 42% of the corpus, though the remaining 58% of studies distribute across a diverse range of complementary mechanisms, reflecting the breadth of approaches the field is exploring [3, 10, 5].

The cross-reference matrix (Fig. 8) reveals structural patterns in the relationship between AI techniques and risk types. Traditional machine learning dominates across the broadest risk portfolio, reflecting its versatility for structured operational data. Deep Learning and hybrid architectures concentrate on disruption, logistics, and demand risks, which are characterised by temporal complexity and high data volume. Generative AI and Knowledge Graphs are emerging in fraud and integrity risk contexts, where the capacity to reason over unstructured information and relational knowledge is particularly valuable. XAI techniques are distributed across supplier, pandemic, and geopolitical risks, confirming that high-stakes, low-frequency risk decisions create the strongest demand for interpretable AI outputs.

The dominance of logistics and transport risk and disruption risk in the cross-reference matrix, relative to the limited coverage of cybersecurity, geopolitical, and fraud risks, reveals a field that is strongly responsive to operationally visible and historically recent disruptions, while exhibiting limited engagement with risks that are either diffuse, novel, or difficult to quantify from structured data alone.

6.2 Emerging Trends

Transition from single-model to multi-agent architectures. The most architecturally ambitious contributions of 2025 deploy systems in which multiple specialised AI components perform distinct functions within a unified pipeline. Rajesh et al. [16] integrate an LLM, Random Forest, K-Means, an LSTM autoencoder, and Reinforcement Learning within a single multi-agent framework. Lin et al. [11] combine Generative AI, Markov modelling, and Mixed Integer Programming. While earlier articles in the corpus already combined multiple techniques [8, 12, 4], the 2025 contributions represent a qualitative escalation in architectural complexity, suggesting a trajectory toward integrated supply chain risk intelligence platforms capable of sensing, reasoning, and acting autonomously.

Entry of Generative AI and LLMs into supply chain risk management. Two 2025 articles deploy Large Language Models as active components of risk management architectures [11, 16], a development with no precedent in earlier articles of the corpus. The capability of LLMs to extract structured risk signals from unstructured text (news, regulatory updates, supplier communications) addresses a longstanding limitation of traditional ML approaches, which require structured input data. This trend is likely to intensify as domain-specific fine-tuning and retrieval-augmented generation techniques lower the cost of deploying LLMs in operational contexts.

Growth of explainability as a design requirement. The increasing prevalence of SHAP and LIME-based interpretability frameworks (evidenced in articles by Adom et al. [1], Kumar and Kumar [8], Li et al. [10], and the systematic review by Nimmy et al. [15]) signals that explainability has transitioned from an optional enhancement to a first-class design requirement. This reflects the practical reality that supply chain risk decisions carry significant financial and operational consequences, requiring not only accurate predictions but transparent justifications that managers can act upon and audit.

Post-COVID-19 acceleration and diversification of risk categories studied. The temporal distribution confirms the COVID-19 pandemic as a research accelerant, generating both the empirical urgency and the data availability needed to develop and validate AI-based risk management systems. Post-2022 articles begin to diversify beyond pandemic risk into geopolitical risk [2], fraud and integrity risk [11], and ESG-related risk [16], suggesting that the field is broadening its risk aperture beyond the immediate disruption context.

6.3 Identified Gaps

The research gaps identified across the corpus are summarised in Fig. 9 and discussed below.

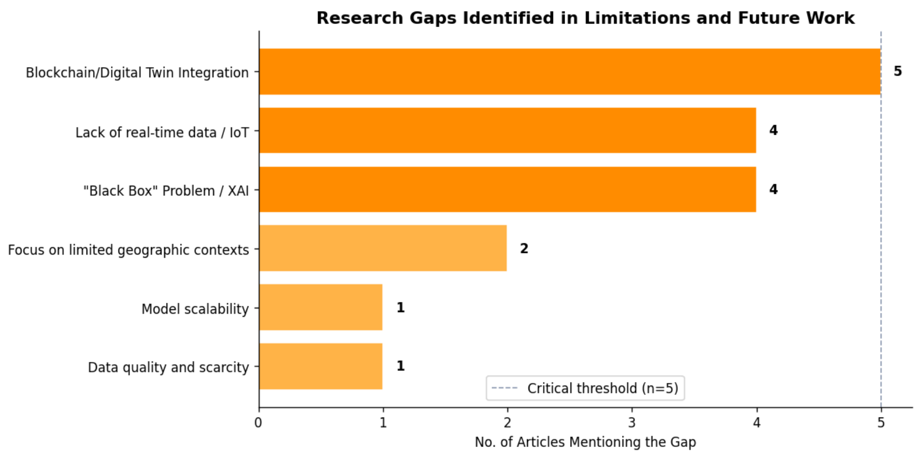

Fig. 9. Research gaps identified across the included articles, ranked by frequency of mention.

Blockchain and Digital Twin integration is the most frequently identified gap, mentioned in five articles. The literature consistently notes that AI-based risk management systems would benefit substantially from integration with blockchain for immutable audit trails and Digital Twin environments for real-time simulation of disruption scenarios [6, 14, 19, 10, 11]. The absence of empirical studies implementing such integrations at operational scale represents the most prominent frontier for future research.

Lack of real-time data and IoT integration is cited in four articles as a barrier to the operationalisation of AI risk management frameworks. Many proposed models are validated on historical batch datasets rather than live operational streams, limiting their applicability for real-time early warning [4, 18, 1, 10]. Future research should prioritise architectures that ingest IoT sensor data, RFID streams, and other real-time supply chain telemetry.

The black box problem and the need for XAI is identified in four articles as a critical barrier to managerial adoption [15, 8, 10, 1]. Despite the growing prevalence of SHAP-based interpretability in recent contributions, the majority of proposed AI systems still lack sufficient transparency for deployment in regulated or safety-critical supply chain environments. Developing explainability frameworks specifically tailored to supply chain risk contexts remains an open research challenge.

Focus on limited geographic contexts is noted in two articles [18, 14]. The concentration of empirical studies in Asian manufacturing and logistics contexts limits the generalisability of findings to supply chains operating under different regulatory frameworks, infrastructure conditions, and risk exposure profiles. Studies conducted in European, North American, and developing-economy contexts are needed to establish the cross-contextual validity of AI-based risk management frameworks.

Model scalability and *data quality and scarcity* are each identified in one article as constraints on the practical deployment of AI systems [10, 3]. Supply chain disruption events are rare and high-impact, creating class-imbalance problems that undermine the reliability of models trained on historical data. Techniques for few-shot learning, synthetic data generation, and transfer learning across supply chain domains represent promising avenues for addressing these constraints.

Absence of cost-benefit analysis for AI implementation. Despite the volume of proposed frameworks and architectures, no article in the corpus quantifies the financial costs, implementation effort, or return on investment associated with deploying AI-based risk management systems in operational supply chain environments. This absence is a meaningful barrier to practitioner adoption: organisations evaluating AI investment decisions require not only evidence of predictive accuracy but also economic justification. Future research should incorporate cost-benefit analysis and total cost of ownership assessments alongside technical performance metrics.

6.4 Limitations of This Review

This review is subject to several limitations that should be considered when interpreting its findings. First, the literature search was restricted to two databases (IEEE Xplore and Web of Science), which, while chosen for their complementary technical and multidisciplinary coverage, may have excluded relevant contributions indexed in Scopus, the ACM Digital Library, or domain-specific repositories. Second, the time restriction to 2019 and 2026 excludes foundational work published prior to 2019 that may underpin more recent methodological contributions. Third, the language restriction to English and Portuguese may introduce a systematic bias against research communities publishing in Mandarin, Arabic, or other languages, which is particularly relevant given the geographic concentration of the corpus in Asia. Finally, the classification of AI techniques and risk types relied in some instances on abstract-level metadata rather than deep full-text algorithmic audits, which may have introduced minor categorisation imprecision.

7 Conclusion

This review mapped the application of artificial intelligence to supply chain risk mitigation across three dimensions the mechanisms deployed, the risk categories addressed, and the techniques employed, drawing on 19 empirical and review studies published between 2022 and 2026. The picture that emerges is one of a field advancing rapidly in technical sophistication, yet still uneven in its coverage of risks, sectors, and geographies.

Early warning and forecasting dominate. About 42% of the corpus is focused on catching disruption signals before they become crises, rather than reacting after damage is done. Traditional machine learning is still the most common technique, mostly because it fits the structured tabular data that ERP and warehouse systems actually produce. The 2025 papers push into hybrid architectures, deep learning combined with graph neural networks, or large language models processing unstructured news feeds, but those remain the exception rather than the norm.

The coverage gaps are substantial. Cybersecurity, geopolitical, and fraud risks barely appear across the nineteen studies. The empirical base is heavily concentrated in Asian manufacturing; European and North American supply chains are almost absent from primary research. Most models are validated on historical batch data, not live operational streams, which raises obvious questions about real-world early warning capability. Interpretability has improved, but not enough for regulated or safety-critical contexts where someone needs to explain a model's output to an auditor or regulator.

The directions for future work follow from these gaps: longitudinal validation in real operating environments; architectures that ingest IoT and real-time transaction data; frameworks built for sectors that have received little attention so far; and empirical studies from geographies other than South and East Asia. The technical foundations exist. What the field still needs to establish is whether they hold up outside the conditions under which they were developed.

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